

Rudresh Veerkhare

MSCS Student, UC San Diego

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Research Interests

My research focuses on **geometry-aware simulation of physical systems**. Right now I'm focused on:

- > **Structure-preserving solvers** — numerical methods for fluids and related systems that respect key physical laws instead of breaking them.
- > **Robust discretization** — turning analytic formulations into mesh-based algorithms that preserve important invariants over time and across resolutions.
- > **Fast prototyping tools** — building *Rheidos*, a framework for quickly experimenting with new simulation and geometry-processing ideas
- > **Learning-augmented simulation** — recently exploring principled ways to use machine learning in physics solvers, rather than as a black-box approximator.

Publications

- [1] **HRescue: A Modern ML approach for Employee Attrition Prediction** [
Rudresh Veerkhare*, Parshwa Shah*, Jiten Sidhpura*, Sudhir Dhage (* = Equal Contribution)
Springer Proceedings in Mathematics & Statistics, vol 401. Springer, Cham. [ICMLBDA 2022]
- [2] **FedSpam: Privacy Preserving SMS Spam Prediction** [
Jiten Sidhpura*, Parshwa Shah*, Rudresh Veerkhare*, Anand Godbole (* = Equal Contribution)
Communications in Computer and Information Science, vol 1793. Springer, Singapore. [ICONIP 2022]
- [3] **Face To BMI: A Deep Learning Based Approach for Computing BMI from Face** [
Jiten Sidhpura*, Rudresh Veerkhare*, Parshwa Shah*, Surekha Dholay (* = Equal Contribution)
In Proceedings of ICITIIT, Kottayam, India, 2022, pp. 1-6. IEEE [ICITIIT 2022]

Work Experience

Present Jun 2025	UC San Diego Center of Visual Computing <i>Graduate Student Researcher Advised by: Prof. Albert Chern</i> Working on structure-preserving point-vortex simulations of incompressible flows on closed surfaces with nontrivial topology using discrete exterior calculus to track global invariants tied to the surface topology and connect them with tools from algebraic geometry and complex analysis .	San Diego, USA
Present Sept 2025	UC San Diego MGTA 402: Data Driven Communications <i>Teaching Assistant Taught by: Kim Walsh</i> Helping students consolidate data, extract insights, and craft effective, data-driven stories .	San Diego, USA
Jun 2025 Mar 2025	UC San Diego CSE 8A: Intro to Programming [ <i>Teaching Assistant Taught by: Prof. Leo Porter</i> Designing exams & quizzes, coordinated programming labs, and conducted office hours & discussion sessions.	San Diego, USA
Mar 2025 Jan 2025	UC San Diego CSE 167: Computer Graphics [ <i>Teaching Assistant Taught by: Prof. Albert Chern</i> Ran weekly office hours for one-on-one help and hour-long discussion sessions on engaging graphics topics; crafted a student-friendly ray-tracing final project ; and worked on a Gradescope autograder for fast, consistent grading.	San Diego, USA
Aug 2024 Jul 2022	Deutsche India Chief Technology Office, TDI <i>Senior Analyst</i> I worked on OCR and Document Understanding to streamline document processing. My work spanned Online Learning Algorithms , Custom Document Entity Extraction , Smart Document Splitting , LLM-aided Contract Drafting , and innovative software automations like Automated Vulnerability Scanning . I have also worked on the adaptation of Document AI within the organization and have experience in Software Governance .	Pune, India

Jul 2021 Jun 2021	Deutsche India Chief Technology Office, TDI Technology Analyst Intern	Pune, India
	As part of the research project, I worked on Optical Character Recognition, which encompassed tasks such as Table Detection , Table Structure Recognition from scanned documents, Signature Extraction , Custom Named Entity Recognition , and Intelligent Character Recognition .	
Apr 2021 Jan 2021	Sardar Patel Institute of Technology Research Assistant Advisor: Prof. Pramod Bide	Mumbai, India
	I conducted research in the field of Cross Event Detection and Topic Mining and Sub-Topic Evolution from Social Media Posts. Developed novel algorithms from scratch using statistical NLP techniques.	

Education

Present Sept 2024	University Of California, San Diego MS, Computer Science & Engineering <i>Specialization: Computer Graphics</i> GPA: 3.73/4	California, USA
May 2022 Aug 2018	Sardar Patel Institute of Technology B.Tech, Computer Engineering CGPA: 9.63/10	Mumbai, India

Awards

- > **India Excellence Award (Deutsche Bank, Feb 2023)** Award granted to only **280 out of 17000** (1.65%).
- > **Best Graduation Project Exhibition Award (SPIT Mumbai, Apr 2022)** Awarded **1st prize** for project **FedSpam**.

Relevant Course Work

- CSE 291: Physics Simulation • CSE 270: Discrete Differential Geometry • CSE 272: Advanced Image Synthesis
- CSE 290: Non-Euclidean Geometry • CSE 276C: Mathematics for Robotics • CSE 203B: Convex Optimization
- CSE 202: Algorithm Design and Analysis

Research Projects

Point Vortex Simulation on Closed Surfaces (solo)

Physics Simulation, Computer Graphics, Discrete Differential Geometry

- > **Structure-preserving** point-vortex solver on 2D surfaces, combining **discrete exterior calculus** with tools from **algebraic geometry**, with a time-stepping scheme that **preserves global invariants** encoded by harmonic 1-forms and vortex positions.

FedSpam: Privacy Preserving SMS Spam Detection (Team Size: 3)

Federated Learning, NLP, Edge Computing

- > FedSpam is an edge computing application which leverages **Federated ML** to **preserve the user data privacy** while using the advanced data-driven ML solutions for spam detection.

CustomXGBoost: XGBoost Implementation with Optimizers (Solo) [[in](#) | [Q](#)]

Gradient Boosting, Optimizers

- > XGBoost **Implementation from scratch** where I've modified the **gradient boosting** to utilize optimizers such as **ADAM** and **RMSProp**.

Arbit: A Decentralized Crypto Exchange Arbitrage System (Solo)

Linear Algebra & Graphs, Blockchain Smart Contracts

- > Developed an efficient algorithm with **$O(1)$** time complexity for detecting **Nth order arbitrage** opportunity in Decentralized Crypto Exchanges (DeX).
- > Utilized graphs and linear algebra in the derivation process.
- > Implemented the system on Cloud with infrastructure provisioned for ensuring low network latency, enabling continuous real-time blockchain monitoring and swift execution of profitable arbitrage transactions.
- > Created a profitable personal project, generating approximately \$10,000 in cryptocurrency in early 2022.

HRescue: A Modern ML approach for Employee Attrition Prediction (Team Size: 3)

Gradient Boosting, Explainable AI, Data Augmentations

- > Numerous attrition prediction methods have been developed in the past. However, This approach focuses on **interpretability** of ML models for sensitive employee attrition decisions and **outperforms prior methods** while addressing **data imbalance**.

Face To BMI: A Deep Learning Based Approach for Computing BMI from Face (Team Size: 3)

Computer Vision, Transfer Learning, Discriminative Learning

- > Developed a BMI prediction model from facial images using **transfer learning** on deep convolution networks, implementing **discriminative learning** to train the last few layers with varying learning rates for further model fine-tuning.
- > Used **Tensor Processing Unit** for training Deep Learning models.

Technical Skill

Programming	C++, Python 3, Java, C, JavaScript, Typescript, Solidity, PowerShell, Shell Script
Frameworks & Libraries	PyTorch, Numpy, Taichi-Lang, Keras, Tensorflow, Scikit-Learn, Huggingface, LangChain, ReactJS, NextJS, Django, Flask, FastAPI, SpringBoot
Tools & Technologies	Git, Houdini by SideFX, Google Cloud Platform (GCP), Amazon Web Services (AWS)

Software Projects

Rheidos (Solo) []

Computer Graphics, Procedural Generation, GPU Computing, Rendering

- > Python-centric framework for **fast prototyping** physics simulations using discrete differential geometry (DDG) with **Taichi-Lang** compute kernels and **Panda3D** rendering, supporting both **interactive Jupyter-driven workflows** (live Panda3D viewport) and scripted demos that configure views/controllers and run in Panda3D's main loop.

Catalyst (Solo) [| |]

Open Source, Node js, VSCode Extension

- > Catalyst is a VS code Extension to accelerate the process of solving problems on Codeforces. It has **4000+ users and 12000+ downloads**

ReactPy (Solo) [|]

Open Source, Web Python, Algorithm

- > ReactPy is a implementation of React in Python using Brython. It's a **from scratch implementation** of **React Fiber**, along with **diffing and Virtual DOM** in Python 3.

Numras (Solo) []

Open Source, Numpy, Neural Networks

- > A **mini-framework** for neural networks implemented in Numpy with a **Keras**-style API, featuring a simple reverse-mode automatic differentiation engine for forward/backward passes and optimization.

Elliptical Curve Diffie Hellman (Solo) []

Cryptography, Algorithms

- > Implemented the **Elliptic Curve Diffie-Hellman (ECDH) key exchange** protocol, including finite-field arithmetic and elliptic-curve point operations.

Volunteering

Mentor, Graduate Women in Computing (GradWIC)

Sept 2025 - Present

- > Provide 1:1 mentoring and peer support to graduate students in computing, sharing guidance on academics, research, and navigating the program.

40+ Hours of CSR, Deutsche India

2022 - 2024

- > Volunteered for development of applications to spread awareness about Mental Health.
- > Volunteered for School Kit Assembly and Distribution for underprivileged children.
- > Volunteered for crafting of environment friendly paper bags.

- › Implemented an Enhancements Proposal for allowing Minkowski distance with $0 < p < 1$.

SEVA, Mumbai

2019 - 2020

- › Volunteered for teaching Maths and Science to high school underprivileged children.
- › Volunteered for Mumbai's Beach Cleaning.

Hackathons

- › **Predicting House Prices In Bengaluru (Machine Hack, Feb 2021)** Ranked **6th** out of 403 submissions (Top 1.4%).
- › **Predict The Data Scientists Salary In India (Machine Hack, July 2020)** Ranked **3rd** out of 192 submissions (Top 1.5%).
- › **Predict The Price Of Books (Machine Hack, July 2020)** Ranked **46th** out of 847 submissions (Top 5%).
- › **Video Game Sales Prediction (Machine Hack, June 2020)** Ranked **24th** out of 231 submissions (Top 10%).
- › **Computer Vision Classic (Machine Hack, July 2020)** Ranked **10th** out of 87 submissions (Top 11%).
- › **JanataHack: Machine Learning for IoT (Analytics Vidya, May 2020)** Ranked **28th** out of 202 submissions (Top 14%).
- › **KJSCE HACK 6.0 (KJSCE Mumbai, Apr 2022)** Won the Filecoin Track Prize of **\$260**.
- › **SPIT Hackathon 2021 (SPIT Mumbai, Feb 2021)** Won the Best Hack Build on Ethereum + Matic Prize of **\$200**.
- › **HackNITR 3.0 (NIT Rourkela, Oct 2021)** Won the Best Dapp Built on Celo Prize of **\$265**.

Online Certifications

- › **Advanced Machine Learning and Signal Processing (IBM, April 2020)** 🔗
 - › **Neural Networks and Deep Learning (Deeplearning.ai, May 2020)** 🔗
 - › **Structuring Machine Learning Projects (Deeplearning.ai, May 2020)** 🔗
 - › **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Deeplearning.ai, May 2020)** 🔗
 - › **Convolutional Neural Networks (Deeplearning.ai, June 2020)** 🔗
 - › **Sequence Models (Deeplearning.ai, July 2020)** 🔗
 - › **Image Super Resolution Using Autoencoders in Keras (Coursera, July 2020)** 🔗
 - › **Generate Synthetic Images with DCGANs in Keras (Coursera, July 2020)** 🔗
 - › **Regression with Automatic Differentiation in TensorFlow (Coursera, July 2020)** 🔗
 - › **Sequences, Time Series and Prediction (Deeplearning.ai, March 2021)** 🔗
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